

1. Administrative & Study Identification

Full Study Title: Prospective Multicentric Cohort Study of Severe and very Severe COPD patients in Brazil **Short Title/Acronym:** SCOPE (Study That Will Evaluate the Brazilian Population With COPD) **Protocol Number:** D5980R00117 **NCT Number:** NCT06780046 **Sponsor Name:** AstraZeneca **Investigational Product:** N/A (Non-interventional, observational cohort study) **Phase of Development:** N/A (Observational) **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the Brazilian population with severe and very severe Chronic Obstructive Pulmonary Disease (COPD) by describing their clinical characteristics, exacerbation rates, progression of the disease, and the occurrence of complications and death. **Secondary Objectives:** To evaluate the impact of different regional risk factors (tobacco smoke, biomass burning, environmental pollution) on disease progression and identify inflammatory biomarkers that correlate with COPD severity and exacerbations.

2.2 Background & Rationale The severity of COPD is intrinsically linked to the exacerbation rate, which drives the rapid progression of the disease, complications, and mortality. Currently, there is a lack of national epidemiological data on this specific subgroup (severe and very severe COPD) in Brazil. Because the diverse regions of Brazil present unique exposure profiles to risk factors, varying population compositions, and different levels of urbanization and education, this prospective multicenter study is needed to generate representative real-world data to better understand COPD outcomes in this population.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Observational, prospective, multicenter cohort, non-interventional **Control Type:** Single-arm cohort (No control) **Blinding:** Open-label **Randomization:** N/A (Non-random consecutive inclusion)

3.2 Planned Enrollment & Duration **Target N\$:** Consecutive enrollment (Sample size scales to capture regional representation) **Number of Sites:** 8 participating centers representing all 5 regions of Brazil (North, Northeast, Midwest,

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Patients of both sexes, aged ≥ 18 years, with confirmed COPD. 2. Diagnosis according to GOLD 2024 guidelines: previous or current exposure and spirometry (FEV_1/FVC ratio and $CAT \geq 10$). 3. Disease severity classified as GOLD E (≥ 2 moderate or 1 severe exacerbation requiring hospitalization); OR GOLD 3 and 4 ($FEV_1 < 50\%$).

4.2 Exclusion Criteria 1. Advanced fibrosing interstitial lung disease (extent $\geq 50\%$). 2. High-risk pulmonary hypertension (currently on triple therapy). 3. Active cancer - currently receiving systemic therapy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: N/A (Observational design; patients receive standard of care according to GOLD 2024 guidelines). **Route of Administration:** N/A **Duration of Treatment:** Patients will be followed prospectively for 12 months.

5.2 Concomitant & Prohibited Therapy Permitted: Standard inhaled maintenance therapies for COPD and treatments for any comorbidities as prescribed by their managing physician. **Prohibited:** N/A (Real-world observational setting).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Baseline (V0)	Day 0	Consecutive Inclusion, Informed Consent, Baseline Clinical & Biomarker Labs
Tele-consult 1	Month 3	Remote gathering of clinical data, exacerbation assessment
Interim	Month 6	On-site Visit: Efficacy assessment, biomarker collection, exacerbation tracking
Tele-consult 2	Month 9	Remote gathering of clinical data, exacerbation assessment
End of Study	Month 12	Final On-site Visit: Clinical assessment, final data capture, exit

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Rate of COPD exacerbations, progression of the disease, and incidence of complications or death over the 12-month follow-up period. **Measurement Method:** Clinical evaluation, CAT/CAAT (COPD Assessment Test) questionnaires, and documented medical/hospitalization records.

7.2 Secondary & Exploratory Endpoints 1. Correlation of specific inflammatory biomarkers with disease progression in the severe COPD cohort. 2. Clinical outcome variations mapped to regional environmental risk exposures (e.g., biomass burning vs. tobacco smoke).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring during standard care, specifically capturing severe and moderate exacerbations via Month 3/9 tele-consults and Month 6/12 on-site visits. **Serious Adverse Events (SAEs):** COPD exacerbations resulting in hospitalization and all-cause mortality recorded as they occur. **Data Monitoring Committee (DMC):** Supervised by AstraZeneca's observational study and real-world evidence governance teams.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Full Analysis Set (FAS) comprising all eligible severe and very severe COPD patients consecutively enrolled. **Sample Size Justification:** Sized to provide statistical significance across the 5 distinct geographical macro-regions of Brazil to ensure an accurate national epidemiological profile. **Handling of Missing Data:** Standard imputation methods applicable to real-world observational cohort data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP), standard real-world data collection practices, and the Declaration of Helsinki. **Monitoring Plan:** Scheduled remote and on-site monitoring to verify source data from the 8 participating medical centers. **Audit Trail:** Robust eCRF locking mechanisms mapping directly to patient hospital files and tele-consultation logs.

11. Study Identifiers & Governance

Primary ID: D5980R00117 **Registry Number:** NCT06780046 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** SCOPE **Official Regulatory Title:** Prospective Multicentric Cohort Study of Severe and very Severe COPD patients in Brazil

12. Clinical Framework (COPD Specific)

Primary Condition: Chronic Obstructive Pulmonary Disease (COPD)
Diagnosis Threshold: Severe / Very Severe per GOLD 2024 (GOLD E or GOLD 3/4 with \$FEV1 < 50\%\$)
Medical Methodology: Standard of Care (SoC) / GOLD Guideline-directed inhaled therapies
Optimization Target: Reduction in annualized moderate-to-severe exacerbation rates

13. Study Procedures & Methodology

Management Pathway: Observational tracking of standard clinical pathways across Brazilian health centers
Education Protocol (HCP): Registry reporting and standardized CAT/CAAT administration training
Education Protocol (Patient): Standard disease management counseling
Quality Audit Mechanism: Remote data capture audits and tele-consult compliance tracking

14. Observation & Follow-Up Schedule

Total Duration: 12 Follow-up months per patient
Onsite Visit Cadence: Baseline (V0), Month 6, Month 12
Remote Monitoring Frequency: Tele-consults at Month 3 and Month 9
Checklist Review Interval: Semi-annual clinical review

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				